Dallas College · School of Engineering, Technology, Math and Science · Physical Sciences Department

University Physics I PHYS-2425

Summer 2026 Section 2 4 Credits 06/08/2026 to 07/09/2026 Modified 06/10/2026

Course Information

Class Meetings:

Type	Location	Days & Times
In-Person Lecture	Richland Campus; Wichita Building; ROOM WH121	Mon/Tues/Wed/Thurs ; 5:00 PM - 7:00 PM
In-Person Laboratory	Richland Campus; Sabine Hall; ROOM SH251	Mon/Tues/Wed/Thurs ; 7:10 PM - 9:10 PM

Withdraw Date : 06/25/2026
Certification Date : 06/11/2026

Course Description

The first semester of calculus - based physics sequence for science, computer science, and engineering majors. The principles and applications of classical mechanics, including harmonic motion, physical systems and thermodynamics are studied with emphasis on problem solving. Performance of basic laboratory experiments supporting theoretical physics principles and applications of classical mechanics, including harmonic motion, physical systems and thermodynamics. Also includes experimental design, data collection and analysis, and preparation of laboratory reports.

Requisites

Student has completed all of the following course(s): MATH 2413 - Calculus I

State-Defined Learning Outcomes

Student Learning Outcomes- Lecture

Upon successful completion of this course, students will:

1. Determine the components of linear motion (displacement, velocity, and acceleration), and especially motion under conditions of constant acceleration.
2. Solve problems involving forces and work.
3. Apply Newton's laws to physical problems. Identify the different types of energy.

4. Solve problems using principles of conservation of energy.
5. Define the principles of impulse, momentum, and collisions.
6. Use principles of impulse and momentum to solve problems.
7. Determine the location of the center of mass and center of rotation for rigid bodies in motion.
8. Discuss rotational kinematics and dynamics and the relationship between linear and rotational motion.
9. Solve problems involving rotational and linear motion.
10. Define equilibrium, including the different types of equilibrium.
11. Discuss simple harmonic motion and its application to real-world problems.

Student Learning Outcomes- Lab

Laboratory activities will support the theoretical principles and students will:

1. Prepare laboratory reports that clearly communicate experimental information in a logical and scientific manner.
2. Conduct basic laboratory experiments involving classical mechanics.
3. Relate physical observations and measurements involving classical mechanics to theoretical principles.
4. Evaluate the accuracy of physical measurements and the potential sources of error in the measurements.
5. Design fundamental experiments involving principles of classical mechanics.
6. Identify appropriate sources of information for conducting laboratory experiments involving classical mechanics.

Instructor-Defined Learning Outcomes

Texas Core Objectives

The College defines essential knowledge and skills that students need to develop during their college experience. These general education competencies parallel the Texas Core Objectives for Student Learning. In this course, the activities you engage in will give you the opportunity to practice two or more of the following core competencies:

A. Critical Thinking Skills: to include creative thinking, innovation, inquiry, and analysis, evaluation and synthesis of information;
B. Communication Skills: to include effective development, interpretation and expression of ideas through written, oral and visual communication;
C. Empirical and Quantitative Skills: to include the manipulation and analysis of numerical data or observable facts resulting in informed conclusions;
D. Teamwork: to include the ability to consider different points of view and to work effectively with others to support a shared purpose or goal;
E. Personal Responsibility: to include the ability to connect choices, actions and consequences to ethical decision-making; and
F. Social Responsibility: to include intercultural competence, knowledge of civic responsibility, and the ability to engage effectively in regional, national, and global communities.

Required Course Materials

[Lecture Materials Link (https://www.bkstr.com/webApp/discoverShop?merfnbr=612&termDir=2026SU&division1=&department1=PHYS&course1=2425&section1=2)](https://www.bkstr.com/webApp/discoverShop?merfnbr=612&termDir=2026SU&division1=&department1=PHYS&course1=2425&section1=2)
[Laboratory Materials Link (https://www.bkstr.com/webApp/discoverShop?merfnbr=612&termDir=2026SU&division1=&department1=PHYS&course1=2425&section1=L2)](https://www.bkstr.com/webApp/discoverShop?merfnbr=612&termDir=2026SU&division1=&department1=PHYS&course1=2425&section1=L2)

Graded Work

The "**Criteria**" table below is a summary of all the graded work in this course.

The "**Breakdown**" table explains the final letter grade.

Criteria

Type	Weight	Topic	Notes
Homework	20%		Homework is through Achieve McMillan. Although you may certainly work with a study group, the ability to work homework problems is essential. In order to be prepared for Exams, make certain you understand how to work each homework problem.
Lab Preparation, Attendance, and Participation	10%		Attending and participating in labs (in-person). Each lab has a pre-lab which must be completed before the beginning of lab and turned in at the beginning of lab. There is a 33% deduction for a missing or incomplete pre-lab. Before you leave lab, your group must complete the lab with results acceptable to your instructor. Students will work together with lab partners and all students must participate, record data and perform calculations. Any students not participating will receive a 33% deduction. Lab will start on-time. Late arrivals may be required to perform any missed portions of the lab themselves as you are not allowed to simply copy data for part of the lab that you missed. No phone usage is allowed during lab. If you need to take care of a personal matter, please step outside. Students must earn a minimum of 60% of total lab assignment points to receive a passing grade in this course. If a student does not earn this minimum, the overall grade will be changed to the average lab grade. If a student earns 60% or greater for the total lab assignment average grade, the student will receive the overall grade as their grade.

Type	Weight	Topic	Notes
Lab Reports	10%		The instructor-provided lab report template must be used for each required lab report. Lab Reports submitted other than on the lab report template will not be graded. Each lab report must be a single file and submitted into ecampus. All calculations must be worked out by hand with all work shown. Students must earn a minimum of 60% of total lab assignment points to receive a passing grade in this course. If a student does not earn this minimum, the overall grade will be changed to the average lab grade. If a student earns 60% or greater for the total lab assignment average grade, the student will receive the overall grade as their grade.
Exams	40%		2 Exams will be given in the semester.
Comprehensive Final Exam	20%		Comprehensive Final Exam.

Breakdown

Grade	Range	Notes
A	90-100	
B	80-89	
C	70-79	
D	60-69	
F	Below 60	

Course Schedule

The table below is a summary of course topics and due dates.

Your instructor will notify you of any changes to the schedule during the term.

When	Topic	Notes

When	Topic	Notes
Lab/Lecture Day 1 - June 8, 2026 Lecture and Lab rooms	Volume 1 Chapter 1 Units and Measurement and Volume 1 Chapter 2 Vectors and Lab. . Lab Introduction.	HW due on 6/10/26 for both chapters. Required safety training. Lab Introduction. Measurement, units and significant digit requirements. Review of Lab Guide. Review of Lab Preparation/Attendance/Participation requirements including pre-lab requirements. Review of Lab Report Requirements.
Lab/Lecture Day 2 - June 9, 2026 Lecture and Lab Rooms	Volume 1 Chapter 3 Motion Along a Straight Line. Lab 1 Measurements.	HW due on 6/11/26. Lab 1 Measurements. The pre-lab is located in the lab manual. Turn in the answers to the pre-lab on paper at the beginning of lab. Lab Report due on ecampus on 6/15/26. The lab report must be written on the lab report template.
Lab/Lecture Day 3 - June 10, 2026 Lecture and Lab Rooms	Volume 1 Chapter 4 Motion in 2 and 3 Dimensions and Lab 2.	HW due on 6/12/26. Lab: Linear Motion: Measuring g Value.. The pre-lab is located in the lab manual. Turn in the answers to the pre-lab on paper at the beginning of lab. Lab Report due on 6/15/26. Must be submitted on ecampus and must be on the lab report template.
Lab/Lecture Day 4 - June 11, 2026 Lecture and Lab Rooms	Volume 1 Chapter 5 Newton's Laws of Motion and Lab 3	HW due on 6/13/26. Lab 3: Force Table. The pre-lab is located in the lab manual. Turn in the answers to the pre-lab on paper at the beginning of lab. Lab Report due on 6/15/26. Must be submitted on ecampus and must be on the lab report template.
Lab/Lecture Day 5 - June 15, 2026 Lecture and Lab Rooms	Volume 1 Chapter 6 Applications of Newton's Laws. Lab 4.	HW due on 6/17/26. Lab 4: Projectile Motion. The pre-lab is located in the lab manual. Turn in the answers to the pre-lab on paper at the beginning of lab. Lab Report due on 6/22/26. Must be submitted on ecampus and must be on the lab report template.
Lab/Lecture Day 6 - June 16, 2026 Lecture and Lab Rooms	Volume 1 Chapter 7 Work and Kinetic Energy. Lab 5.	HW due on 6/18/25. Lab 5: Dynamics of a Rolling Cart. The pre-lab is located in the lab manual. Turn in the answers to the pre-lab on paper at the beginning of lab. Lab Report due on 6/22/26. Must be submitted on ecampus and must be on the lab report template.
Lab/Lecture Day 7 - June 17, 2026 Lecture and Lab Rooms	Volume 1 Chapter 8 Potential Energy and Conservation of Energy. Lab 6.	HW due on 6/22/26 due to the holiday. Lab 6: Centripetal Motion. The pre-lab is located in the lab manual. Turn in the answers to the pre-lab on paper at the beginning of lab. Lab Report due on 6/22/26. Must be submitted on ecampus and must be on the lab report template.

When	Topic	Notes
Lab/Lecture Day 8 - June 18, 2026 Lecture and Lab Rooms	Volume 1 Chapter 9 Linear Momentum and Collisions. Lab 7	HW due on 6/22/25 due to the holiday. Lab 7: Kinetic Energy and Potential Energy. The pre-lab is located in the lab manual. Turn in the answers to the pre-lab on paper at the beginning of lab. Lab Report due on 6/22/26. Must be submitted on ecampus and must be on the lab report template.
Lab/Lecture Day 9 - June 22, 2026 Lecture and Lab Rooms	Volume 1 Chapter 10 Fixed-Axis Rotation. Tutoring during Lab.	HW due on 6/24/26. Tutoring during lab.
Lab/Lecture Day 10 - June 23, 2026 Lecture and Lab Rooms	Exam Volume 1 Chapters 1-8 during lecture. Lab 8.	You may have 1 hand-written sheet of paper (front and back) with equations and constants only. If you arrive with more than 1, you must choose which one to use. It must be hand-written. You must have a calculator which is not on your phone, tablet, computer, etc. Lab 8: Impulse and Momentum. The pre-lab is located in the lab manual. Turn in the answers to the pre-lab on paper at the beginning of lab. Lab Report due on 6/29/26. Must be submitted on ecampus and must be on the lab report
Lab/Lecture Day 11 - June 24, 2026 Lecture and Lab Rooms	Volume 1 Chapter 11 Angular Momentum. Lab 9	HW due on 6/26/26. Lab 9: Dynamic Carts/Collisions and Momentum, The pre-lab is located in the lab manual. Turn in the answers to the pre-lab on paper at the beginning of lab. Lab Report due on 6/29/26. Must be submitted on ecampus and must be on the lab report.
Lab Day 12 - June 25, 2026 Lecture and Lab Rooms	Volume 1 Chapter 12 Static Equilibrium and Elasticity. Lab 10.	HW due on 6/27/26. Lab 10: Ballistic Pendulum. The pre-lab is located in the lab manual. Turn in the answers to the pre-lab on paper at the beginning of lab. Lab Report due on 6/29/26. Must be submitted on ecampus and must be on the lab report.
Lab/Lecture Day 13 - June 29, 2026 Lecture and Lab Rooms	Volume 1 Chapter 13 Gravitation. Lab 11.	HW due on 7/1/26. Lab 11: Rotational Inertia of a Wheel. The pre-lab is located in the lab manual. Turn in the answers to the pre-lab on paper at the beginning of lab. Lab Report due on 7/6/26. Must be submitted on ecampus and must be on the lab report.
Lab/Lecture Day 14 - June 30, 2026 Lecture and Lab Rooms	Volume 1 Chapter 15 Oscillations. Lab 12.	HW due on 7/2/26. Lab 12: Simple Harmonic Motion. The pre-lab is located in the lab manual. Turn in the answers to the pre-lab on paper at the beginning of lab. Lab Report due on 7/6/26. Must be submitted on ecampus and must be on the lab report.

When	Topic	Notes
Lab/Lecture Day 15 - July 1, 2026 Lecture and Lab Rooms	Volume 2 Chapters 1 and 2 Temperature and Heat and The Kinetic Theory of Gases. Lab 13.	HW for both chapters is due on 7/6/26 due to the holiday. Lab 13: Latent Heat and Specific Heat. The pre-lab is located in the lab manual. Turn in the answers to the pre-lab on paper at the beginning of lab. Lab Report due on 7/6/26. Must be submitted on ecampus and must be on the lab report.
Lab/Lecture Day 16 - July 2, 2026 Lecture and Lab Rooms	Volume 2 Chapters 3 and 4 The First and Second Laws of Thermodynamics. Lab 14.	HW for both chapters is due on 7/6/26 due to the holiday. Lab 14: Mechanical Equivalent of Heat. Turn in the answers to the pre-lab on paper at the beginning of lab. Lab Report due on 7/6/26. Must be submitted on ecampus and must be on the lab report.
Lab/Lecture Day 17 - July 6, 2026 Lecture and Lab Rooms	Review during lecture. Tutoring during lab.	Review during lecture. Tutoring during lab.
Lab/Lecture Day 18 - July 7, 2026 Lecture and Lab Rooms	Exam Volume 1 Chapters 9-13, 15 and Volume 2 Chapters 1-4 during lecture. Tutoring during lab.	You may have 1 hand-written sheet of paper (front and back) with equations and constants only. If you arrive with more than 1, you must choose which one to use. It must be hand-written. You must have a calculator which is not on your phone, tablet, computer, etc. Tutoring during Lab time.
Lab/Lecture Day 19 - July 8, 2026 Lecture and Lab Rooms	Review during lecture. Tutoring during lab.	Review during lecture. Tutoring during lab.
Final Day 20 - July 9, 2026 Lecture room	Comprehensive Final Exam during lecture time.	You may have 1 hand-written sheet of paper (front and back) with equations and constants only. If you arrive with more than 1, you must choose which one to use. It must be hand-written. You must have a calculator which is not on your phone, tablet, computer, etc.

* Course Policies

Tutoring.

Other than the lab sessions specified for tutoring, tutoring is available by appointment.

Unexpected Class Changes

If there are unexpected class changes, they will be posted in announcements on ecampus. If there are any Dallas College notifications, they will be communicated through media and the Dallas College website.

Attendance and Participation

In order to be successful students must attend and participate in enrolled courses. There is no make-up for missed work.

A grade is associated with lab attendance.

Late Work

No late homework or lab reports will be accepted. One of each will be dropped.

Missed exams cannot be made-up except when a College acceptable excuse (i.e. illness warranting a physician's care, death in the immediate family, religious absences, and sanctioned college athlete's events) is supplied with documentation through email prior to the exam.

No work may be submitted for additional credit after the final exam.

Support Contacts

Contact Your Success Coach (https://www.dallascollege.edu/successcoach)
Every Dallas College student has a personalized Success Coach who supports them from day one to graduation. Contact your coach for help navigating college and reaching milestones leading to graduation and a career.

Get Free Tutoring (https://www.dallascollege.edu/tutoring)
Tutoring is free to all current Dallas College students. You can walk in or schedule an appointment at all Learning Commons (https://www.dallascollege.edu/learningcommons) campus locations. Live, online tutoring is also available via eCampus.

Explore More Free Student Resources (https://www.dallascollege.edu/help)
You have access to many free resources as a Dallas College student, including Counseling and Psychological Services (https://www.dallascollege.edu/counseling), Child Care Resources (https://www.dallascollege.edu/childcare), Housing Resources (https://www.dallascollege.edu/housing), Emergency Aid Funds (https://www.dallascollege.edu/emergencyaid), Food Pantries (https://www.dallascollege.edu/foodpantry), and more!

Submit the Student Care Form (https://www.dallascollege.edu/careform)
Not sure which free college resources can help? Submit the Student Care Form! Our Student Care Network (https://www.dallascollege.edu/studentcare) will connect you to support for physical and mental health, financial concerns, food, clothing, and more.

Contact Technical Support (https://www.dallascollege.edu/techsupport)
Need help with eCampus or another college technology? Our technical support staff can assist you.

Institutional Policies

Dallas College Policies

Please review the **Institutional Policies** (https://www.dallascollege.edu/syllabipolicies) page to learn about accommodations for students with disabilities, class drop and repeat options, Title IX (harassment, discrimination, and sexual misconduct), and more.